

Branches of Logic and Reference Works

Logic has developed many specialized subfields throughout history. Major branches include: **Aristotelian (syllogistic) logic**, **Boolean logic**, **Buddhist logic**, **Bunched logic**, **Categorical (algebraic) logic**, **Classical (mathematical) logic**, **Computability logic**, **Deontic logic**, **Dependence logic**, **Deviant logic**, **Doxastic logic**, **Epistemic logic**, **First-order (predicate) logic**, **Free logic**, **Fuzzy logic**, **Higher-order logic**, **Infinitary logic**, **Informal logic**, **Intensional logic**, **Intermediate logic**, **Interpretability logic**, **Intuitionistic logic**, **Linear logic**, **Many-valued logic**, **Metalogic**, **Minimal logic**, **Modal logic**, **Non-Aristotelian logic**, **Non-classical logic**, **Non-monotonic logic**, **Ordered logic**, **Paraconsistent logic**, **Philosophical logic**, **Provability logic**, **Quantum logic**, **Relevance logic**, **Substructural logic**, **Temporal logic**, **Term (syllogistic) logic**, **Topical (dialectical) logic**, and others ¹ ². Each branch has its own foundational literature; below we highlight two standard references for each branch (in any language) along with historical context where relevant:

- **Aristotelian (syllogistic) logic** – The ancient logic of terms and syllogisms. Standard references include Parry & Hacker's *Aristotelian Logic* (SUNY Press, 1991) ³ and Jonathan Barnes's Oxford translation of Aristotle's *Posterior Analytics* (2nd ed. 1994) ⁴. (Aristotle's *Prior Analytics* is similarly collected in Ross (ed.) 1951 ⁵.)
- **Boolean logic** – The algebra of truth values introduced by George Boole. Foundational works include Boole's *An Investigation of the Laws of Thought* (1854) ⁶ and, for modern treatment, Halmos's *Lectures on Boolean Algebras* (1963) ⁷, which develop Boolean algebra and logic. (Huntington's *Boolean Algebra* and Stone's representation theorems are also classic.)
- **Buddhist logic** – The logical and epistemological traditions in Indian Buddhism. A canonical reference is F. Th. Stcherbatsky's *Buddhist Logic* (Vols. I–II, Dover/PMLA, 1930/1962) ⁸, which presents the classical Indian treatises by Dignāga and Dharmakīrti in translation and commentary. (Modern studies include works on Navya-Nyāya logic and Tibetan logic.)
- **Bunched Logic** – A substructural logic for reasoning about resource combination (e.g. the *logic of Bunched Implications*, BI). The foundational paper is Peter O'Hearn & David Pym, "The Logic of Bunched Implications" (Bulletin of Symbolic Logic 5(2), 1999), which introduced BI and separation logic. Recent expositions include O'Hearn's lectures on separation logic and the edited volume *Contemporary Trends in Logic* (eds. Gabbay et al., 2015, see also [47]).
- **Categorical Logic** – The study of logic via category theory and topos theory. A standard monograph is Bart Jacobs's *Categorical Logic and Type Theory* (Elsevier, 1999) ⁹. (See also Goldblatt, *Topoi: The Categorical Analysis of Logic* (1979).) Jacobs's book develops the internal logic of categories and cartesian closed categories.
- **Classical (Mathematical) Logic** – The traditional logic of Boolean connectives and quantifiers. Standard textbooks include Herbert Enderton's *A Mathematical Introduction to Logic* (2nd ed. 1972)

¹⁰ and Elliott Mendelson's *Introduction to Mathematical Logic* (3rd ed. 1987) ¹¹, which cover propositional, first-order, and completeness/compactness theorems.

- **Computability Logic** – A modern formalism introduced by Giorgi Japaridze (2003) viewing logic as interactive computability. The foundational paper is Japaridze's "*Introduction to Computability Logic*" (Annals of Pure & Applied Logic 123(1–3):1–99, 2003). See also Japaridze's later works on game semantics and resource-conscious logics.
- **Deontic Logic** – Logic of obligation, permission, and norm. A key anthology is *Handbook of Deontic Logic and Normative Systems* (ed. van der Torre & Meyer, 2009/2013) ¹², which surveys modern deontic systems. Early collections by R. Hilpinen include *Deontic Logic: Introductory and Systematic Readings* (Reidel, 1971) ¹³.
- **Dependence Logic** – Logics of informational dependence (team semantics). Introduced by Jouko Väänänen, cf. *Dependence Logic: A New Approach to IF-Logic* (Cambridge Univ. Press, 2007) and developed further in the edited volume *Dependence Logic: Theory and Applications* (eds. Abramsky et al., Springer, 2017). (Väänänen's work is cited as a "new area of logic introduced by Väänänen" ¹⁴.)
- **Deviant Logic** – A term (coined by Susan Haack) for non-classical logics. Susan Haack's *Deviant Logic* (1987) introduced critical analysis of alternative logics; it was later reprinted (with new material) as *Deviant Logic, Fuzzy Logic: Beyond the Formalism* (1996) ¹⁵. Haack's work examines "the philosophical consequences of alternative logics" beyond classical logic ¹⁵.
- **Doxastic Logic** – Modal logic of belief. Classic sources include Hintikka's *Knowledge and Belief: An Introduction to the Logic of the Two Notions* (Cornell U. Press, 1962) ¹⁶, and the multi-agent logic text Fagin, Halpern, Moses & Vardi *Reasoning About Knowledge* (MIT Press, 1995) ¹⁷, which, while focused on knowledge, also addresses belief modalities.
- **Epistemic Logic** – Modal logic of knowledge (and often belief). Foundational works are also Hintikka (1962) ¹⁶ and Fagin et al. (1995) ¹⁷ as above. Additional references include Meyer's papers and works in the *Handbook of Epistemic Logic*.
- **First-Order (Predicate) Logic** – Extension of propositional logic with quantifiers. Refer to Enderton (1972) ¹⁰ and Mendelson (1987) ¹¹ for proofs of completeness, compactness, and basics of model theory (Chang & Keisler's *Model Theory* (1990) is a classic advanced text).
- **Free Logic** – Predicate logic without existential import (allowing empty names). Standard references include Karel Lambert's *Free Logics: Their Foundations, Character, and Some Applications* (Academia Verlag, 1997) ¹⁸, and *New Essays in Free Logic* (eds. Morscher & Hieke, CSLI Publ. 2001) ¹⁹, which honor Lambert's work.
- **Fuzzy Logic** – Logic with graded truth values in $[0,1]$ (Zadeh's fuzzy sets). A seminal reference is Petr Hájek's *Metamathematics of Fuzzy Logic* (Kluwer, 1998) ²⁰. For comprehensive introductions see Cintula, Hájek & Noguera (eds.), *Handbook of Mathematical Fuzzy Logic* (2 vols., 2011, 2015).

- **Higher-Order Logic** – Logic allowing quantification over predicates/functions. Foundational treatments include Church's *Introduction to Mathematical Logic* and Henkin's completeness for second-order logic. (Automated proof environments often cite e.g. *Automath* or *HOL* manuals.)
- **Infinitary Logic** – Logics permitting infinite disjunctions/conjunctions (e.g. $L_{\omega_1\omega}$). Chang & Keisler's *Model Theory* covers such logics; also *Infinitary Logic: In Memoriam Carol Karp* (Lecture Notes, 1975) contains essays on the subject.
- **Informal Logic** – The study of everyday reasoning and argumentation. Key texts are Douglas Walton's *Informal Logic: A Pragmatic Approach* (Cambridge U.P., 2008) ²¹ and R. H. Johnson & J. A. Blair's *Logical Self-Defense* (2nd ed., 1977), which survey fallacies, argument schemes, and critical thinking.
- **Intensional Logic** – Logic of intensional contexts (e.g. modalities, propositional attitudes). Classic works include Carnap's *Meaning and Necessity* (1956) ²² and Hughes & Cresswell's *A New Introduction to Modal Logic* (1968) ²³. Gallin's *Intensional and Higher-Order Modal Logic* (North-Holland, 1975) is another standard source.
- **Intermediate Logic** – Logics strictly between classical and intuitionistic logic (often studied via extensions of intuitionistic propositional calculus). A reference is D. Bostock's *Intermediate Logic* (2nd ed., 2001). (Some are covered in *Handbook of Philosophical Logic* vol. 7 by Font & van Benthem.)
- **Interpretability Logic** – Modal logic encoding interpretability relations between theories. Key literature includes Visser's and de Rijke's papers (e.g. in *Handbook of Proof Theory*, 1998). (See also Beklemishev's works on GLP modal logic.)
- **Intuitionistic Logic** – Logic of constructive truth (Brouwer–Heyting logic). Seminal sources are Martin Dummett's *The Logical Basis of Intuitionistic Mathematics* (1977) ²⁴ and van Dalen's edition of Brouwer's *Cambridge Lectures on Intuitionism* (1981) ²⁵. For proof theory see Troelstra & van Dalen *Constructivism in Mathematics* (1988).
- **Linear Logic** – A resource-sensitive substructural logic introduced by Jean-Yves Girard. Girard's original paper "*Linear Logic*" (*Theoretical Computer Science* 50:1–102, 1987) introduced the system ²⁶. Important references include Girard's *Collected Papers (1987, 1998)* and the edited volume *Advances in Linear Logic* (eds. Girard, Lafont & Regnier, 1995) ²⁷.
- **Many-Valued Logic** – Logics with more than two truth values. A comprehensive treatment is in Cignoli, D'Ottaviano & Mundici's *Algebraic Foundations of Many-Valued Reasoning* (Trends in Logic, Vol. 7, 2000) ²⁸. Also Siegfried Gottwald's *A Treatise on Many-Valued Logics* (3 vols., 2001) ²⁹ surveys historical and algebraic developments.
- **Mathematical Logic / Metalogic** – General study of formal logical systems. Standard references are Enderton (1972) and Mendelson (1987) as above, and also *Computability and Logic* by Boolos & Jeffrey (4th ed., 2002). (Meta-topics like completeness and decidability are central here.)

- **Minimal Logic** – The weakest normal logic without ex falso. See the introduction in e.g. *Basic Proof Theory* by Troelstra & Schwichtenberg.
- **Modal Logic** – Logic of necessity and possibility. Essential references include Blackburn, de Rijke & Venema's *Modal Logic* (Cambridge, 2001), Hughes & Cresswell (1996), and Ruth Barcan Marcus's *Modalities* (OUP, 1993) ³⁰. (The *Handbook of Modal Logic* vol. 3 also collects many articles.)
- **Non-Aristotelian / Non-Classical Logics** – This umbrella term covers deviations from classical logic. See Haack's critiques (above) and works on non-classical logics generally, such as Dunn & Hardegree's *Algebraic Methods in Philosophical Logic* (2001) ³¹. Paraconsistent, relevance, quantum, fuzzy etc. fall under this category.
- **Non-Monotonic Logic** – Logics of defeasible or default reasoning (withdrawable inferences). The foundational paper is Reiter's "*A Logic for Default Reasoning*" (AI Journal 13:81–132, 1980) ³². Collections by Ginsberg (1987) and the Stanford Encyclopedia (2019) survey default logic, autoepistemic logic, circumscription, etc ³³ ³². (For example, Ginsberg's "*Handbook of Non-Monotonic Reasoning*" has key articles.)
- **Ordered Logic** – Logics imposing order constraints on sequents. (Less common; see papers on ordered logic for sequential resource reasoning.)
- **Paraconsistent Logic** – Logics that tolerate contradictions without triviality. Seminal works include Graham Priest's *In Contradiction* (2nd ed. 2006) and the anthology *Paraconsistent Logic: Essays on the Inconsistent* (ed. Priest et al., Munich, 1989). (Also relevant: Routley–Meyer semantics and Belnap's *First-Degree Entailment*.)
- **Philosophical Logic** – The application of logic to philosophical problems (e.g. propositional attitudes, vagueness, etc.). Key references are in *Handbook of Philosophical Logic* (multiple volumes, ed. Gabbay & Guenther, 1984–present), covering topics like tense logic, belief logic, semantics, etc.
- **Provability Logic** – Modal logic of formal provability in arithmetic. The classical text is Boolos's *The Logic of Provability* (Cambridge, 1993) ³⁴. See also Artemov & Beklemishev (2004) in the *Handbook of Philosophical Logic* ³⁵ and Smoryński's *Self-Reference and Modal Logic* (1991) for the arithmetical perspective.
- **Quantum Logic** – The logic of quantum propositions (projectors on a Hilbert space forming an orthomodular lattice). The founding paper is Birkhoff & von Neumann "*The Logic of Quantum Mechanics*" (Annals of Math. 37:823–843, 1936) ³⁶. Garrett Birkhoff's *Lattice Theory* (2nd ed., AMS, 1967) ³⁶ and Příkrý's *Quantum Logics* (trans. 1971) are further references.
- **Relevance Logic** – Logics requiring antecedent and consequent of implications to share relevant terms. The classic systems were introduced by Anderson & Belnap (*Entailment*, 1975). A modern source is Dunn & Restall's entry "Relevance Logic" in *Handbook of Philosophical Logic*, Vol. 6 (2002) ³⁷.

- **Substructural Logic** – Logics that drop structural rules (exchange, weakening, contraction) from Gentzen systems. Covers linear, relevance, Lambek calculus, etc. See Paoli's *Substructural Logics* (Oxford U.P., 2001) and Dunn & Hardegree *Algebraic Methods in Philosophical Logic* (2001) ³¹.
- **Temporal Logic** – Logic of time (with operators like "always" and "eventually"). Standard references are Manna & Pnueli's *The Temporal Logic of Reactive and Concurrent Systems* (Springer, 1992) ³⁸ and Emerson's "Temporal and Modal Logic" chapters in *Handbook of Theoretical Computer Science* (1990). Pnueli's introduction of LTL (1977) and Kröger & Merz's books are also key.
- **Term (Syllogistic) Logic** – Essentially Aristotelian logic of terms and syllogisms (identified with "traditional logic"). Foundational texts are Aristotle's *Prior Analytics* and *Posterior Analytics*. Modern editions include Ross's critical edition (1951) ⁵ and Robin Smith's translations (*Prior Analytics*, 1989 ³⁹).
- **Topical (Dialectical) Logic** – Logic of dialectical argument ("topics" or loci of argumentation from rhetoric). Traces back to Aristotle's *Topics*. The standard edition is Robin Smith's *Aristotle, Topics I & VIII and Selections* (Clarendon, 1997) ³⁹. (Topical logic was developed in late antiquity, see e.g. van der Lugt's works on Aristotle's Topics.)

(For each of the above branches, numerous additional monographs and articles exist. The citations above point to representative texts in English.)

Influential Papers and Monographs

Some landmark works that have profoundly shaped logic and related science include:

- **Principia Mathematica** (Whitehead & Russell, 1910–13) – monumental formalization of logic (no citation).
- **Gödel (1931)** – "Über formal unentscheidbare Sätze", showing incompleteness of formal arithmetic.
- **Turing (1936)** – "On Computable Numbers", laying foundations of computation theory.
- **Church (1936)** – proof that the Entscheidungsproblem is undecidable.
- **Hilbert–Ackermann (1928)** – early formal logic textbook introducing decision problems.
- **Löwenheim–Skolem (1915–20)** – showing downward Löwenheim–Skolem theorem (Skolem paradox).
- **Tarski (1936)** – definition of truth and model theory.
- **Cantor (1891)** – diagonal argument showing uncountability of the reals.
- **Brouwer (1913)** – intuitionism and rejection of classical logic.
- **Curry (1942)** – paradox in combinatory logic (precursor of Curry's paradox).
- **Beth (1956)** – semantic tableaux method.

(Each of these historic works is documented in secondary sources on logic's history.)

History, Philosophy, and Applications

Historical/Philosophical Works: Important books on the history and philosophy of logic include Florian Cajori's *A History of Mathematics* (1909, contains early logic), Imre Lakatos's *Proofs and Refutations*, Ivor

Grattan-Guinness's *The Search for Mathematical Roots* (4 vols, 1990s), and Michael Dummett's *Elements of Intuitionism* (1977). Stewart Shapiro's *Philosophy of Mathematics: Structure and Ontology* (2014) and Graham Priest's *Towards Non-Being* (2005) discuss logical pluralism and paradox. The multi-volume *Handbook of Logic and Foundations* (ed. van Heijenoort & Gerner, 1992) surveys many topics.

Applications of Logic: Books showing logic in action include Ebbinghaus et al., *Mathematical Logic* (Springer, 3rd ed. 1984), Enderton's *Elements of Set Theory*, and computer-science texts like Huth & Ryan's *Logic in Computer Science*. Philosophical applications are in textbooks like Priest's *Introduction to Non-Classical Logic* (2nd ed., 2008). Logic also underpins areas like formal linguistics (Lukasiewicz & Tarski, Montague grammar) and AI (Brachman & Levesque's *Knowledge Representation and Reasoning*).

Cultural Enrichment and Popular Logic

Logic also pervades culture and puzzles. Notable examples include:

- **Lewis Carroll (C. L. Dodgson)** – His *Symbolic Logic* (1896) and *Games in Logic* (1887) present logical ideas and puzzles.
- **Raymond Smullyan** – Author of many popular logic puzzle books (e.g. *What Is the Name of This Book?*).
- **Knights and Knaves puzzles** – Logical puzzles invented by Smullyan.
- **Gödel, Escher, Bach** by Douglas Hofstadter (1979) – Explores self-reference and logic in art/mathematics.
- **Martin Gardner's "Mathematical Games"** columns – Introduced many logic puzzles.
- **Magic squares, Sudokus, and brainteasers** – Recreational problems based on logical constraints.
- **C.S. Lewis's *The Lion, the Witch and the Wardrobe*** – Contains logic paradox (the Dwarfs' forest rule).
- **Prominent Logicians** – Anecdotes (e.g. Wittgenstein's logic notebooks, John von Neumann's brainteasers).

These works and activities, while not formal logic, demonstrate logical thinking in an engaging, cultural context.

Sources: Branch names and historical context are drawn from logic outlines ¹ ² . Book references are from publishers' descriptions and bibliographies ³ ⁴ ⁶ ⁷ ⁴⁰ ⁸ ⁹ ⁴¹ ¹² ¹³ ¹⁴ ¹⁵ ¹⁶ ¹⁷ ¹⁸ ¹⁹ ²⁰ ²² ²³ ⁴² ²⁶ ²⁸ ²⁹ ³⁰ ⁴³ ³⁶ ³⁷ ³⁸ ³⁹ , among others. Each cited work exemplifies the given branch.

¹ ² Outline of logic - Wikipedia

https://en.wikipedia.org/wiki/Outline_of_logic

³ Aristotelian Logic - William Thomas Parry, Edward A. Hacker - Google Books

https://books.google.com/books/about/Aristotelian_Logic.html?id=rjceFowdGEAC

⁴ ⁵ ³⁹ Aristotle's Logic (Stanford Encyclopedia of Philosophy)

<https://plato.stanford.edu/entries/aristotle-logic/>

- 6 George Boole, An Investigation of the Laws of Thought - PhilPapers
<https://philpapers.org/rec/BOOAIO>
- 7 31 Boolean algebra - Wikipedia
https://en.wikipedia.org/wiki/Boolean_algebra
- 8 Buddhist Logic - Fedor Ippolitovich Shcherbatskoï - Google Books
https://books.google.com/books/about/Buddhist_Logic.html?id=NFdTwAEACAAJ
- 9 Type theory - HandWiki
https://handwiki.org/wiki/Type_theory
- 10 11 41 Classical Logic (Stanford Encyclopedia of Philosophy/Fall 2017 Edition)
<https://plato.stanford.edu/archivES/FALL2017/entries/logic-classical/>
- 12 13 Deontic Logic (Stanford Encyclopedia of Philosophy)
<https://plato.stanford.edu/entries/logic-deontic/>
- 14 Dependence Logic: Theory and Applications | SpringerLink
https://link.springer.com/book/10.1007/978-3-319-31803-5?error=cookies_not_supported&code=d0025c7b-1fdf-4f74-ad90-263a3776a0e3
- 15 Amazon.com: Deviant Logic, Fuzzy Logic: Beyond the Formalism: 9780226311340: Susan Haack: Books
<https://www.amazon.com/Deviant-Logic-Fuzzy-Beyond-Formalism/dp/0226311341>
- 16 Knowledge and belief : an introduction to the logic of the two notions : Hintikka, Jaakko, 1929-2015 : Free Download, Borrow, and Streaming : Internet Archive
https://archive.org/details/knowledgebeliefi00hint_0
- 17 Reasoning About Knowledge
<https://mitpress.mit.edu/9780262562003/reasoning-about-knowledge/>
- 18 19 Free Logic (Stanford Encyclopedia of Philosophy)
<https://plato.stanford.edu/entries/logic-free/>
- 20 28 29 Fuzzy Logic (Stanford Encyclopedia of Philosophy)
<https://plato.stanford.edu/entries/logic-fuzzy/>
- 21 Douglas N. Walton, Informal Logic: A Pragmatic Approach - PhilPapers
<https://philpapers.org/rec/WALILA-6>
- 22 23 Intensional Logic (Stanford Encyclopedia of Philosophy)
<https://plato.stanford.edu/entries/logic-intensional/>
- 24 25 42 Intuitionistic Logic (Stanford Encyclopedia of Philosophy)
<https://plato.stanford.edu/entries/logic-intuitionistic/>
- 26 27 Linear Logic (Stanford Encyclopedia of Philosophy)
<https://plato.stanford.edu/entries/logic-linear/>
- 30 Modal logic - Wikipedia
https://en.wikipedia.org/wiki/Modal_logic
- 32 33 Non-monotonic Logic (Stanford Encyclopedia of Philosophy)
<https://plato.stanford.edu/entries/logic-nonmonotonic/>

34 35 43 Provability Logic (Stanford Encyclopedia of Philosophy)

<http://plato.stanford.edu/entries/logic-provability/>

36 Quantum Logic and Probability Theory (Stanford Encyclopedia of Philosophy)

<https://plato.stanford.edu/entries/qt-quantlog/>

37 Relevance Logic (Stanford Encyclopedia of Philosophy/Winter 2018 Edition)

<https://plato.stanford.edu/archives/win2018/entries/logic-relevance/>

38 The Temporal Logic of Reactive and Concurrent Systems: Specification | SpringerLink

<https://link.springer.com/book/10.1007/978-1-4612-0931-7>

40 Buddhist Logic Vol.i : Stcherbatsky, F. Th. : Free Download, Borrow, and Streaming : Internet Archive

<https://archive.org/details/in.ernet.dli.2015.533476>